

Question	Marking Guidance	Mark	Comments
09.1	1. (DNA) helicase causes breaking of hydrogen/H bonds (between DNA strands); 2. DNA polymerase joins the (DNA) <u>nucleotides</u>; 3. Forming <u>phosphodiester</u> bonds;	3	1. Reject 'helicase hydrolyses hydrogen bonds'. 2. Reject if suggestion that DNA polymerase joins the complementary nucleotides or forms H bonds. 2. Reject if joining RNA nucleotides or forming RNA.

Question	Marking Guidance	Mark	Comments
09.2	1. (Treatment D Antibody binds to cyclin A so) it cannot bind to DNA/enzyme/initiate DNA replication; 2. (Treatment E) RNA interferes with mRNA/tRNA/ribosome/polypeptide formation (so cyclin A not made); 3. In Treatment F added cyclin A can bind to DNA/enzyme (to initiate DNA replication) OR Treatment F shows that it is the cyclin A that is being affected in the other treatments OR Treatment F shows that cyclin A allows the enzyme to bind (to DNA) OR (Some cells in D or E) can continue with DNA replication because they have a different cyclin A <u>allele</u> OR (Some cells in D or E) can continue with DNA replication because the antibody/RNA has not bound to all the cyclin A protein/mRNA OR (Some cells in E) can continue with DNA replication because they contain previously translated cyclin A;	3	1. For 'bind to enzyme' accept 'activate'. 1. Idea of 'initiate DNA replication' must be linked to start not just less replication. 1. and 3. For 'enzyme' accept named enzyme. 3. Context needed for Treatment F but it does not need to be named.

